

SCROLLING FOUR-DIGIT SEVEN-SEGMENT DISPLAY

Project Documentation

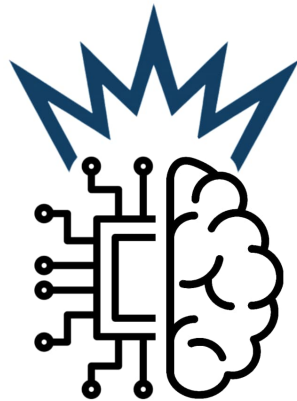
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Digital Design Using Verilog HDL

Abstract

This project presents the design and implementation of a scrolling four-digit seven-segment display using Verilog HDL. The system displays a sequence of hexadecimal values on a four-digit seven-segment display. Clock division is used to generate slower clock signals from the main input clock. A scrolling controller changes the displayed values at a slow rate, while multiplexing is used to control the four digits of the display.

The project demonstrates important digital design concepts such as clock division, counters, multiplexing, seven-segment decoding, and sequential control logic. The design is suitable for implementation and testing on an FPGA development board.

1 Introduction

Seven-segment displays are commonly used in digital systems to display numbers and simple characters. A multi-digit seven-segment display contains multiple individual digits that share segment connections.

In this project, a four-digit seven-segment display is controlled using Verilog HDL. The system generates suitable clock signals, changes the displayed values according to a pre-defined scrolling sequence, and selects each digit one after another using multiplexing.

The rapid switching of the digits makes all four digits appear continuously illuminated to the human eye.

2 Project Overview

The project consists of the following major functions:

- Clock division
- Scrolling sequence generation
- Four-digit multiplexing
- Seven-segment decoding
- Display control

The main input clock is divided into slower clock signals. One clock is used for controlling the scrolling operation, while another is used for rapidly switching between the four display digits.

2.1 Objectives

The main objectives of the project are:

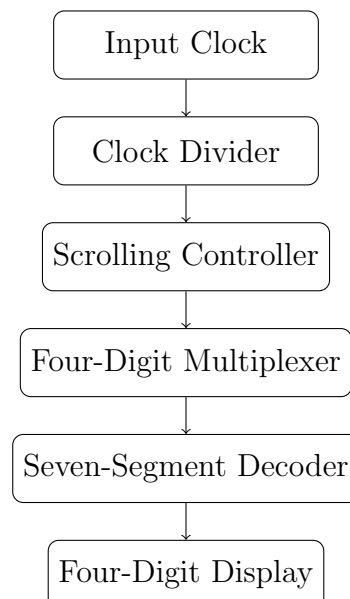
- To design a four-digit seven-segment display using Verilog HDL.
- To generate slower clock signals using clock dividers.
- To implement a scrolling sequence.
- To multiplex four seven-segment digits.
- To decode numerical values into seven-segment patterns.
- To verify the design using simulation.
- To implement the design on an FPGA board.

3 System Architecture

The system can be divided into four main blocks:

1. Clock Divider
2. Scrolling Controller
3. Digit Multiplexer
4. Seven-Segment Decoder

The overall operation can be represented as follows:



3.1 Clock Division

The FPGA provides a high-frequency clock signal. Directly using this clock for scrolling would make the display change too quickly to be observed.

Therefore, counters are used to divide the input clock frequency and generate slower clock signals.

Two different clock rates are used:

- A slower clock for the scrolling operation.
- A faster clock for display multiplexing.

The slower clock controls how frequently the displayed sequence changes, while the faster clock allows the four digits to be switched rapidly.

3.2 Scrolling Sequence Controller

The scrolling controller determines which four values are displayed at each step.

A counter is used to move through the scrolling positions. The counter cycles through the predefined sequence and then repeats.

The scrolling sequence contains six positions:

Position	Digit 1	Digit 2	Digit 3	Digit 4
0	2	2	1	0
1	3	2	1	1
2	0	3	2	2
3	1	0	3	2
4	2	1	0	3
5	2	2	1	0

After the final position, the sequence starts again from the first position.

3.3 Four-Digit Multiplexing

The four-digit seven-segment display shares the segment lines between the digits. Therefore, only one digit is activated at a time.

A digit-selection counter cycles through four positions:

$$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$$

At each position, the corresponding digit-select line is activated and the appropriate segment pattern is applied.

Because the switching occurs rapidly, the human eye perceives all four digits as being continuously ON.

3.4 Seven-Segment Decoder

The seven-segment decoder converts the numerical value into the corresponding segment pattern.

For this project, the values 0, 1, 2, and 3 are displayed.

Input Value	Displayed Character
0	0
1	1
2	2
3	3

The decoder produces the required signals for the seven-segment display.

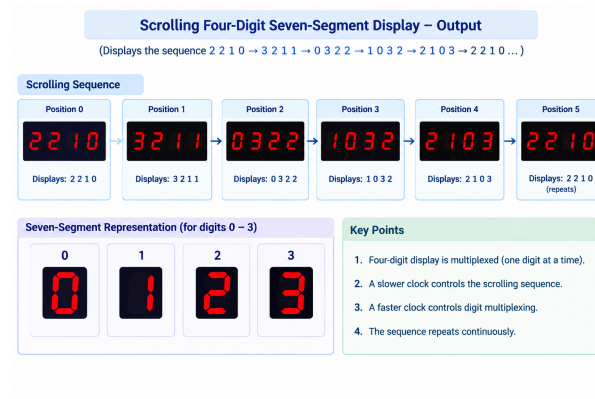
4 System Operation

The complete operation of the system takes place in the following sequence:

1. The FPGA receives the input clock.
2. The clock divider generates slower clock signals.
3. The scrolling controller uses the slower clock to change the displayed sequence.
4. Four values are assigned to the four display positions.
5. The multiplexing counter selects one digit at a time.
6. The selected value is sent to the seven-segment decoder.
7. The decoder generates the required segment pattern.
8. The process repeats continuously.

5 Flow Chart

The operation of the project can be represented by the following flow chart.



6 RTL Design

The RTL design shows the hardware structure generated from the Verilog HDL design.

The major hardware functions include clock division, scrolling control, digit selection, and seven-segment decoding.

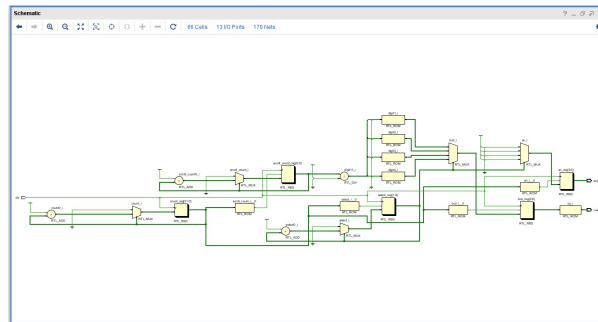


Figure 1: RTL design of the scrolling seven-segment display

7 Simulation Results

The simulation verifies the correct operation of the scrolling controller and display control logic.

The waveform demonstrates the changing scrolling position, digit selection, and corresponding seven-segment output signals.

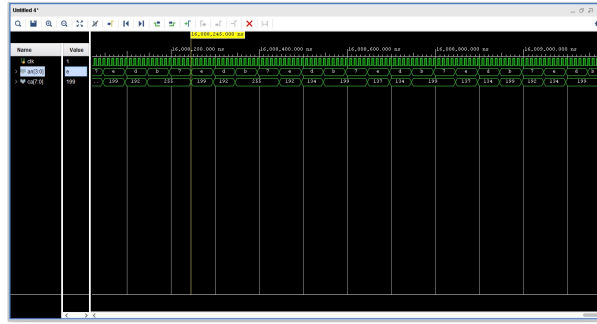


Figure 2: Simulation waveform of the scrolling seven-segment display

8 Results

The design successfully performs the following operations:

- Generates slower clock signals from the input clock.
- Produces the required scrolling sequence.
- Selects the four display digits using multiplexing.
- Generates the appropriate seven-segment patterns.
- Repeats the scrolling sequence continuously.
- Produces the expected simulation waveform.

The RTL design and simulation results confirm the functional operation of the proposed system.

9 Applications

The concepts used in this project can be applied in:

- Digital clocks
- Counters
- Scoreboards
- Digital measuring instruments
- FPGA-based display systems
- Embedded digital systems

10 Advantages

- Simple and easy-to-understand digital architecture.
- Efficient use of a four-digit seven-segment display.
- Demonstrates clock division and multiplexing.
- Suitable for FPGA implementation.
- Easy to modify the scrolling sequence.
- Can be extended for larger display systems.

11 Future Enhancements

The project can be extended by adding:

- More scrolling patterns.
- User-controlled scrolling speed.
- Additional numerical values.
- Alphabetic character display.
- Push-button control.
- Timer or counter functionality.
- More digits using multiplexing.

12 Conclusion

The scrolling four-digit seven-segment display was successfully designed using Verilog HDL. The project demonstrates important digital design concepts including clock division, counters, scrolling control, multiplexing, and seven-segment decoding.

The RTL structure and simulation waveform verify the intended operation of the design. The project also provides a practical understanding of how multiple seven-segment digits can be controlled efficiently using FPGA-based digital logic.